

TAHIA ANAN DHIRA

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EDUCATION

- MSc in Applied Social Data Science (ASDS) (Degree: Distinction) 2023
The London School of Economics and Political Science, London, United Kingdom
Relevant coursework: Social Network Analysis; Quantitative Text Analysis; Applied Machine Learning; Managing and Visualizing Data; Research Design
- MSS in Economics (CGPA: 3.88 / 4.00) 2017
University of Dhaka, Dhaka, Bangladesh
- BSS in Economics (CGPA: 3.94 / 4.00) 2016
University of Dhaka, Dhaka, Bangladesh

RESEARCH INTERESTS

International Trade, Trade Networks, Environmental Economics, Computational Social Science, Applied Econometrics, Network Science

RESEARCH MANUSCRIPTS

Trading Carbon Through Global Supply Chains: Network Structure and the Environmental Kuznets Curve (*working paper, 2026*)

- Compiled and harmonized multi-country datasets on embodied carbon emissions, trade, and macroeconomic indicators.
- Constructed weighted embodied carbon trade networks and computed network statistics for longitudinal analysis. Combined structural gravity modelling, network analysis, and panel econometrics to study embodied carbon trade and environmental outcomes.

Product Reproducibility and International Trade Patterns: Evidence from Cultural Goods (*submitted, 2026*)

- **Constructed and analysed global trade networks** for unique, reproducible, and non-cultural products.
- Estimated structural gravity models (PPML) and dynamic network models (STERGM) to analyse bilateral trade relationships across 190 countries.
- Prepared the manuscript and submitted it as the corresponding author.
- Developed, validated, and debugged reproducible research pipelines using GitHub, Claude Code, and GitHub Copilot for data processing, estimation, and visualization.

Who converges with whom? Global migration network and the geography of development (*working paper, 2026*)

- Constructed a panel of directed weighted bilateral migration networks spanning 23 years.
- Developed reproducible workflows for network construction, visualization, and manuscript figures.
- Contributed to the conceptual framework and manuscript preparation.

Gendered Cultural Hierarchies in Social Media Audience Discourse: Evidence from Bangladeshi Artists (*working paper, 2026*)

- Developed a Python pipeline using the YouTube Data API v3 to collect and process over **8 million** comments.
- Conducted sentiment analysis using **VADER** and topic modelling using the **quanteda** package in R
- Implemented reproducible data processing and analysis workflows in Python and R.

RESEARCH EXPERIENCE

Capstone Project, MS ASDS, LSE | 2023

Formation of friendship network: Analysing assortment of pro-sociality and personality type

- Designed and implemented a behavioural classroom experiment to study homophily and friendship network formation.
- Estimated Exponential Random Graph Models (ERGMs) to identify structural and attribute-based drivers of network formation.
- Examined the role of personality traits, and prosocial behaviour in network formation

Coursework: ST445, MS ASDS, LSE | 2022–2023

Topic-based content and sentiment analysis of COVID-19 vaccine: a study using Reddit

- Contributed to the manuscript preparation and network visualization.

Mental Health of University Students Before and After Online Education

University Grants Commission (UGC)-funded project | 2020–2021

- Contributed to the implementation of a longitudinal survey examining changes in university students' mental health before and after the transition to online education.
- Coordinated data collection, and validated survey instruments using econometric methods.

Willingness to Pay for Treating Mental vs. Physical Illness

Bangladesh Medical Research Council (BMRC)-funded project | 2020–2021

- Contributed to the coordination and implementation of a comparative survey for university students and adults.

PUBLICATIONS

Huque, R., **Dhira, T. A.**, et al. (2023). Determinants of client satisfaction with health insurance: Evidence using a mixed-method approach from Bangladesh. *SSM - Health Systems*, 3. <https://doi.org/10.1016/j.ssmmh.2023.100264>

Rahman, M. A., **Dhira, T. A.**, et al. (2022). PHQ-9 validation in Bangladesh. *PLOS ONE*.

<https://doi.org/10.1371/journal.pone.0269634>

Dhira, T. A., Rahman, M. A., et al. (2021). GAD-7 validation in Bangladesh. *PLOS ONE*.

<https://doi.org/10.1371/journal.pone.0261590>

Mehareen, J., Rahman, M. A., **Dhira, T. A.**, et al. (2021). Mental health during COVID-19. *Journal of Affective Disorders Reports*. <https://doi.org/10.1016/j.jadr.2021.100179>

ACADEMIC APPOINTMENTS

Assistant Professor, Department of Economics, University of Dhaka

October 2025 – present

Lecturer, Department of Economics, University of Dhaka

November 2018 – September 2025

- Taught 10 different undergraduate courses in economics.
- Supervised undergraduate dissertations and research projects.
- Trained STATA to the undergraduate and graduate students.
- Developed computational course materials using R, Python, and Stata.

Lecturer, Department of Economics, East West University

March 2018 – October 2018

TECHNICAL SKILLS

Programming & Data Analysis

- Python (NumPy, Pandas, Matplotlib, NetworkX)
- R (igraph, statnet, ergm, tergm, sna, quanteda, stm, fixest)
- Stata

Methods

- Structural Gravity Models
- Panel Econometrics
- Network analysis

- ERGM & STERGM
- Quantitative Text Analysis (NLP)

Research Workflow & Tools

- Git, GitHub,
- GitHub Copilot,
- Claude Code,
- Code verification, debugging, and validation for AI-assisted programming workflows
- LaTeX (beginner),
- Qualtrics

AWARDS AND SCHOLARSHIP

Commonwealth Master's Scholarship	2022
Multiple Gold Medals for highest CGPA (BSS & MSS Economics)	2016, 2017

LANGUAGE

Advanced fluency in English and Bangla